Milestone 1 Report

SalesGenie AI Platform · Lead Management & Lead Intelligence Engine

Covers SRS Module 1 (Lead Management) and Module 2 (Lead Intelligence) · Stack: FastAPI, Streamlit, PostgreSQL 16, LangGraph

1. Summary

Milestone 1 delivers the first working slice of SalesGenie: sales reps can **add and import leads**, and an AI pipeline **profiles each company, scores it 0-100, and promotes strong leads to Qualified**. The code, API, database migration and tests are in place. What remains is mostly **evidence for sign-off**: a PostgreSQL-backed test run, a sample dataset, ingestion measurements and a formal demo review.

Leads API	Intelligence	Tests	Sign-off
Create + bulk import read, search, stats	**2-agent LangGraph** score, personas, promotion	**22 passed** after 2 fixes, 14.22 s	**Pending** evidence items open

2. Scope and status against the plan

Planned work	Where it lives	Status
Kickoff, requirements, architecture, DB review	SRS PDF, database design, agent design	**Done** (documented)
API contracts, stack, repo setup, sprint plan	API contract, SETUP.md, database SETUP.md	**Done**
ER diagram, PostgreSQL schema, migrations, wireframes, agent architecture	ER diagram, data dictionary, Alembic initial migration	**Done**
Prospect CRUD APIs and company profile analysis	backend/main.py, module1_leads.py, module2_intelligence.py	**Partial**: create/read and analysis done; update/delete not exposed
Lead import, Leads UI, qualification scoring	bulk import service, frontend, scoring routes	**Partial**: in code; ingestion-vector evidence pending
Unit tests for import, CRUD, scoring	test_lead_management.py, test_lead_intelligence.py	**Partial**: tests pass; PostgreSQL integration run pending
Intelligence UI integration and M1 report	intelligence_view.py, this report, test report	**Pending**: formal sign-off

3. System architecture

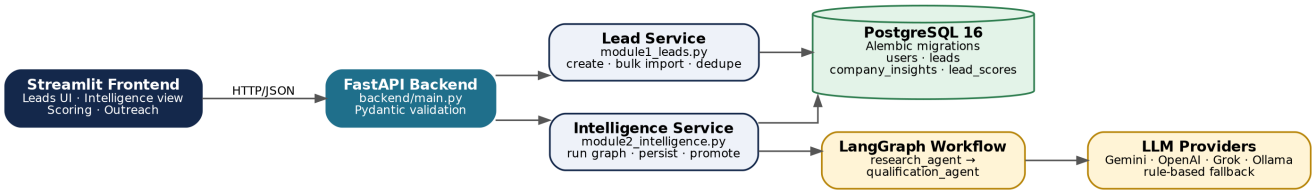


Figure 1: Request path from the Streamlit UI through FastAPI to the services, the LangGraph workflow and PostgreSQL.

The AI workflow is **synchronous**. Each agent tries the configured LLM provider (Gemini, OpenAI, Grok or Ollama) with a bounded timeout and falls back to a deterministic rule-based result if the provider times out or returns incomplete data.

4. Implemented product flow

- A lead is submitted via `POST /api/leads` or `POST /api/leads/bulk`.
- Pydantic validates required fields, email format, source and lifecycle status.

- The lead service checks **case-insensitive email duplicates** and commits the record.
- `GET /api/leads` and `GET /api/leads/stats` expose the ingestion and pipeline view.
- `POST /api/leads/{lead_id}/analyze` runs the research and qualification agents.
- Insight and score are saved to `company_insights` and `lead_scores`; a score of **70 or more** promotes *New Leads* to *Qualified*.
- The frontend consumes the lead, scoring, intelligence and outreach routes.

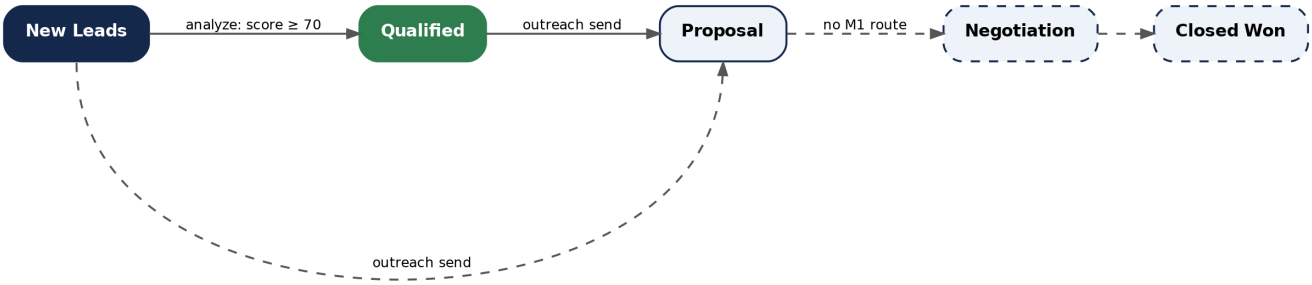


Figure 2: Lead status lifecycle. Dashed paths are valid statuses but are not driven by any Milestone 1 route.

5. Database design

PostgreSQL 16 with schema managed by **Alembic** (do not use `Base.metadata.create_all()`). Milestone 1 uses four tables; *outreach_emails* (M2) and *interactions* (M3) are design entries only and are not in the initial migration.

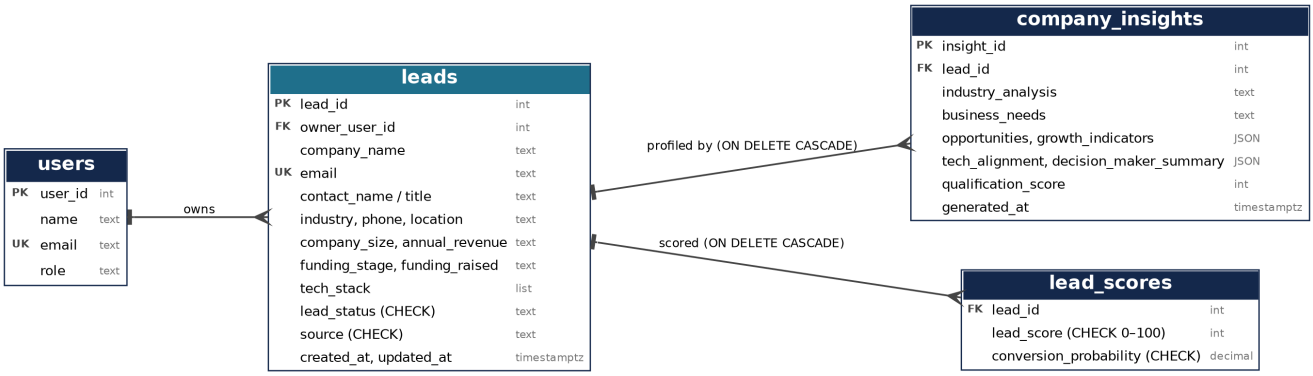


Figure 3: Milestone 1 ER diagram using the live names from the migration and API (partial columns shown for lead_scores).

Table	Purpose	Key rules
users	Seeded sales users; lead owners	Unique email; default users created if table is empty
leads	Prospect profile, contact, source, lifecycle status	Unique email; CHECK on lead_status and source; created_at/updated_at
company_insights	Latest generated company intelligence	FK to leads, ON DELETE CASCADE; JSON columns for structured output
lead_scores	Score and conversion probability	FK to leads, ON DELETE CASCADE; CHECK ranges on score and probability

Allowed values. `lead_status`: New Leads, Qualified, Proposal, Negotiation, Closed Won. `source`: CRM, Website Form, LinkedIn, Sales Entry, CSV. Scores are integers 0-100.

6. Lead Intelligence Engine (Module 2)

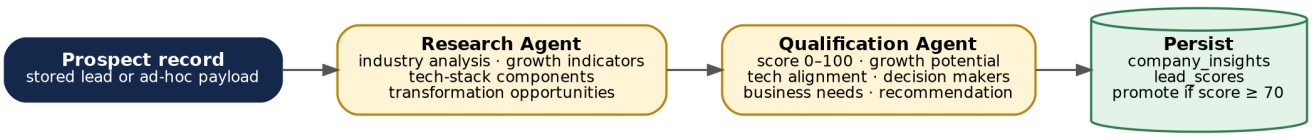


Figure 4: Two-node LangGraph pipeline, research_agent then qualification_agent, followed by persistence.

Agent	Responsibility	Output
Research	Parses domain and tech stack (cloud, backend, frontend, data, DevOps); finds growth indicators and transformation opportunities (ERP modernization, cloud migration, process automation, digital transformation)	industry analysis, growth indicators, tech components, opportunities
Qualification	Weighs funding stage and budget capacity; computes score; maps buying-committee personas; writes recommendation	score 0-100, high-growth flag, tech alignment, decision makers, business needs, recommendation

- **Requirements covered:** REQ-M2-01 company profiling and tech categorization, REQ-M2-02 opportunity identification, REQ-M2-03 intelligence summary, scoring and personas.
- **Conversion probability** is derived as qualification_score x 0.85.
- **Priority bands** in the scoring queue: hot at 75 or above, warm 40-74, cold below 40 (separate from the 70 promotion threshold).
- **Performance target:** end-to-end analysis under 5 seconds per prospect. A fixture test checks this (M1-T10).
- `POST /api/intelligence/analyze-prospect` runs the same pipeline without saving anything.

7. API summary

Base URL `http://localhost:8000`. No authentication yet, and CORS currently allows all origins, so restrict both before production. The generated OpenAPI schema (`/docs`, `/openapi.json`) is the machine-readable source of truth.

Method	Path	Purpose	OK
GET	/health	API availability	200
GET	/api/users	List seeded sales users	200
POST	/api/leads	Create one lead	201
POST	/api/leads/bulk	Import many leads; skip_duplicates default true	201
GET	/api/leads	Search/list leads (search, source, limit 1-200)	200
GET	/api/leads/stats	Total leads, ingestion health, pending scoring	200
POST	/api/leads/{id}/analyze	Run agents, persist insight and score, promote	200
GET	/api/leads/{id}/insights	Latest persisted insight (404 if none)	200
POST	/api/leads/batch-analyze	Analyze selected leads; per-item errors	200
POST	/api/intelligence/analyze-prospect	Ad-hoc analysis, nothing saved	200
GET	/api/intelligence/architecture	LangGraph topology	200
GET	/api/scoring/queue	All leads with latest score or unscored	200
GET	/api/leads/{id}/dossier	Profile + score + insights (generates if missing)	200
POST	/api/leads/{id}/score	Re-run and persist scoring	200
POST	/api/leads/score-batch	Score every unscored lead	200
GET	/api/llm/providers	Provider availability	200

Errors: 400 duplicate or domain error, 404 unknown lead/insight, 422 schema validation, 503 AI provider unavailable or incomplete output. Application errors return `{"detail": "message"}`. The outreach routes (`/api/outreach/*`) also exist in the API but belong to Milestone 2 scope; save-draft and send only acknowledge, and send moves a lead from New Leads or Qualified to Proposal.

CRUD status

Operation	State in Milestone 1
Create (single and bulk)	**Done**. Duplicate returns 400 (strict bulk mode aborts with no partial insert).
Read / list / search / stats	**Done**
Update (PUT / PATCH)	**Pending**: not exposed. Do not build clients against it yet.
Delete	**Pending**: not exposed. The database already cascades insight and score rows.

8. Testing

Focused run after installing `requirements-dev.txt` and `requirements-ai.txt`: **22 passed, 1 warning in 14.22 s**. The warning is a deprecation notice from the Google GenAI dependency and does not fail any test.

Two failures found and fixed

- **SLA breach:** two sequential, unbounded provider calls exceeded 5 seconds. Gemini calls now have a bounded timeout and both agents fall back to rule-based results.
- **Outreach output:** responses could omit the company name or fail on provider timeout. The formatter now enforces company context and a local draft fallback exists.

Milestone 1 test cases

ID	Area	Expected result
M1-T01	Single lead creation	Valid lead mapped to a Lead record and committed
M1-T02	Single lead duplicate	Existing email raises a validation error; nothing committed
M1-T03	Bulk import	Existing and in-batch duplicate emails skipped by default
M1-T04	Bulk strict mode	A duplicate aborts the import with no partial commit
M1-T05	Request validation	Invalid email and unsupported status rejected
M1-T06	Agent compilation	Graph available with research and qualification nodes
M1-T07	Agent profiling	Industry analysis and categorized tech components returned
M1-T08	Opportunity detection	ERP, cloud and automation opportunities found for the fixture
M1-T09	Qualification output	Score, business needs, recommendation, decision makers returned
M1-T10	Latency	End-to-end fixture completes in under 5 seconds

The lead-management tests cover service behaviour without a live database, so schema constraints, foreign keys, timestamps and transactions are not yet verified against real PostgreSQL.

9. Acceptance status

Demonstrated in the repository	Still required for final sign-off
• Lead creation and bulk import behaviour • Duplicate protection and request validation • LangGraph compilation and two-agent execution • Company profiling, opportunity detection, scoring, personas, promotion logic • API contract and migration documentation	• Committed sample dataset under `data/sample` • PostgreSQL-backed schema integrity and transaction test run • Measured ingestion success across all five planned ingestion vectors • Formal UI/demo review and sign-off • Update/delete routes, if full CRUD is required